



National Institute of Technology Kurukshetra
Department of Computer Engineering
End Semester Theory Examination
November / December 2025

Program: B. Tech. CSE A & B and M&C

Semester: III

Subject: Computer Organization and Architecture

Subject Code: CSPC-203

Date of Exam: 02/12/2025

Time: 09:30 AM - 12:30 PM [3 Hours]

Max Marks: 50

Roll No:

Note: All Questions (Five) are compulsory. Each question carries 10 marks.

	(a) Consider a memory system with 1M bytes of main memory and 16K bytes of cache memory. Assume that the processor generates a 20-bit memory address, and the cache block size is 16 bytes. How many bits will be required to store all the tag values in case of Direct mapping, Associative mapping, and 8-way Set Associative mapping? Also, calculate the tag directory size in each case.	6 Marks
Q1.	(b) A computer has a memory hierarchy consisting of a two-level cache (L1 and L2) and a main memory. If the processor needs to access data from memory, it first looks into the L1 cache. If the data is not found in the L1 cache, it is sent to the L2 cache. If it fails to get the data from L2 cache, it goes to main memory, where the data is definitely available. Hit rates and access times of various memory units are shown in the figure. Compute the average memory access time in nanoseconds. <pre> graph TD P[Processor] --> L1[L1 cache Hit rate=95%, Access time = 10 ns] L1 --> L2[L2 cache Hit rate=85%, Access time (including L1 cache miss penalty) = 20 ns] L2 --> MM[Main Memory Access time (including L1 and L2 cache miss penalty) = 200 ns] </pre>	4 Marks
Q2.	(a) (i) A processor has 64 general-purpose registers and 50 distinct instruction types. An instruction is encoded in 32 bits. What is the maximum number of bits that can be used to store the immediate operand for the given instruction? ADD R1, #25 // R1 = R1 + 25 (ii) Consider a memory system with the configuration of 1024 * 8 and identify the size of the data register, address register, program counter, instruction register, temporary register, and accumulator.	3 + 3 Marks
	(b) Consider a disk drive with the specifications: 16 surfaces, 512 tracks/surface, 512 sectors/track, 1 KB/sector, and 3000 rpm with a single R/W head. Whenever a 4 Byte word is available, it is transferred between memory and I/O with a memory cycle time of 40 ns. Calculate the idle time for the processor blocked due to DMA (in %).	4 Marks

Q3.	(a) Given page reference: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. Compare the number of page faults using LRU and FIFO page replacement algorithms by taking 3 frames first and then 4 frames.	6 Marks
	(b) Consider a non-pipelined processor operating at 2.5 GHz. It takes 5 clock cycles to complete an instruction. You are going to make a 5-stage pipeline out of this processor. Overheads associated with pipelining force you to operate the pipelined processor at 2 GHz. In a given program, assume that 30% are memory instructions, 60% are ALU instructions, and the rest are branch instructions. 5% of the memory instructions cause stalls of 50 clock cycles each due to cache misses, and 50% of the branch instructions cause stalls of 2 cycles each. Assume that there are no stalls associated with the execution of ALU instructions. Calculate the speedup achieved by the pipelined processor over the non-pipelined processor.	4 Marks
Q4.	(a) Explain how the expression $X = A * B + C * D$ will be executed in one-address, two-address, and three-address processors in an accumulator organization. Show the complete process in detail.	6 Marks
	(b) A program contains 1000 instructions. Out of that 25% instructions require 4 clock cycles, 40% instructions require 5 clock cycles, and the remaining require 3 clock cycles for execution. Find the total time required to execute the program running on a 1 GHz machine.	4 Marks
Q5.	(a) Explain the design of the ALU in detail with the help of a block diagram.	6 Marks
	(b) Differentiate between combinational circuits and sequential circuits in detail with an example for each.	4 Marks

*****End of Question Paper*****